MIL-I-81454(AS)
AMENDMENT 2
15 JUNE 1970
Superseding
Amendment 1
15 October 1968

Military Specification
Indicator, Gyroscope, Vertical Reference
ID-1481 (*)/A

This Amendment forms a part of a Military Specification MIL-I-81454(AS) dated 15 January 1967 and has been approved by the Naval Air Systems Command, Department of the Navy.

*Page 2, Paragraph 2.1, under Specifications Military: Add

"MIL-L-6085	Lubrication Oil, Aircraft Instrument, Low Volatility
MIL-M-7793	Meter, Time Totalizing
MIL-I-81219	Indicator, Elapsed Time, Electro-Chemical
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-C-26482	Connectors, Electric, Circular, Miniature, Quick Disconnect, Environment Resisting
MIL-G-81322	Grease, Aircraft, General Purpose, Wide Temperature Range
MIL-G-81559	Gyroscope Assemblies, Attitude and Directional Reference Instruments for Aircraft Preservation and Packaging, Procedures for
MIL-B-81744	Barrier Coating Solution and Solvent"

*Page 2, Under Standard, Military: Delete "MIL-STD-794" and Title, Add "MIL-STD-1334 Barrier Coating Process for Anti-Friction Bearings" and "MIL-STD-838 Lubrication of Military Equipment" and "MIL-STD-471 Maintainability Demonstration."

*Page 3, Paragraph 3.1: Delete and substitute

"3.1 Qualification - This specification makes provision for Qualification Testing. (See 6.6)."

*Page 4, Paragraph 3.3.2.3: Delete 1500 and substitute 2000.
Add

"3.3.2.4 Time Totalizing Meter - The indicator shall contain a time totalizing meter (4,000 hours) conforming to Specification MIL-M-7793. The reading of the meter shall not require unsealing of the indicator case. The time totalizer shall be installed within the present envelope of the instrument.

FSC 6610

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:28Z
Check the source to verify that this is the current version before use.

3.3.2.5 Indicator-Elapsed Time, Electro-Chemical - This indicator may be used as an alternate and shall conform to MIL-I-81219 within present envelope of the instrument. The scale shall be for 5,000 hours."

Page 5, Paragraph 3.3.9.1: Delete and Substitute

"3.3.9.1 Vibration - The indicator shall operate satisfactorily when subjected to the vibration tests of 4.8.18 and 4.8.19."

Page 8: Add

"3.5.1.2.5 Drum Mask - The Drum Mask is affixed to rotate with the Roll Gimbal. Referring to Fig. 2, the portion of the Drum Mask above the drum shall be Gray Color No. 36440 of FED-STD-595."

Page 10, Paragraph 3.5.1.7, Last two sentences: Delete and Substitute

"The warning flag shall be fluorescent Red, Color No. 38905 of FED-STD-595 and the lettering "off" shall be Black Color No. 37038 of FED-STD-595. The fluorescent Red flag shall be illuminated under night lighting conditions and the lighting shall not be entirely dependent upon one light source."

Page 11, Paragraph 3.5.1.10: Add

"This device shall not restrict the pitch and roll freedom in the absence of power."

Page 11, Paragraph 3.1.1.11.3, Last Sentence: Delete and Substitute

"The erection rate in pitch and roll shall not exceed 0.9 degree per minute when the spin axis displacement exceeds 10 degrees in pitch and roll."

Page 11, Paragraph 3.5.1.12: Delete and Substitute

"3.5.1.12 Rotor Coast Time - The indicator shall display pitch and roll information within 6 degrees of vertical for a minimum of 9 minutes after power is removed."

*Page 12: Add

"3.6.1 Lubrication

3.6.1.1 Rotor Bearing Lubrication - The gyro spin bearings shall be lubricated with Grease MIL-G-81322 or MINIPURE grease or equivalent with particle size not exceeding 50 MICRONS for grease. The requirements of MIL-STD-838 shall be met.

3.6.1.2 Bearing Lubrication - The lubricating oil used in gimbal bearings shall meet the performance requirements of Specification MIL-L-6085 except that it shall conform with Naval Research Laboratory polyolester lubricant NRL MB-20-B.Teflon separators shall be used in gimbal bearings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:28Z
Check the source to verify that this is the current version before use.

3.6.1.3 Barrier Film Bearing Coating - All bearings lubricated in accordance with requirements of Paragraphs 3.6.1.1 and 3.6.1.2 shall be processed in accordance with MIL-STD-1334. The coating material shall be in accordance with MIL-B-81744. All bearings shall have chamfered races and shall be barrier film processed either by a ball bearing manufacturer or a designated Government facility.

3.7 Operating - Precessions in pitch and roll with rated power shall be in accordance with 3.7.1, 3.7.2 and 3.7.3.

3.7.1 Longitudinal Acceleration - When subjected to 8g acceleration for 4 seconds along the roll axis, the indicator shall not precess in pitch and roll beyond 2 degrees. (Testing of this requirement shall be in accordance with Paragraph 4.8.9.2.3.)

3.7.2 Coordinated Turns - When subjected to coordinated turns with rate of turn between 90 to 1500 degrees per minute and bank angles of 20 to 60 degrees, the indicator shall not precess in pitch and roll beyond 2 degrees maximum. (Testing of this requirement shall be in accordance with Paragraphs 4.8.9.2.1 and 4.8.9.2.2.)

3.7.3 Uncoordinated Turns - When subjected to uncoordinated turns (skid turns) of less than 5 degrees per minute at air speeds not greater than 400 knots for a 10 minute duration, the indicator shall not precess in pitch and roll beyond 2 degrees maximum."

*Page 12, 13, 14 and 15, Paragraphs 4.1.1, 4.2, 4.2.1, 4.2.2, 4.2.3, 4.2.4, 4.3, 4.3.1, 4.3.2, 4.3.3, 4.3.4, 4.4 and 4.4.2: Delete and Substitute

"4.2 Classification of Test - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Qualification Inspection
(2) Quality Conformance Inspection

4.3 Qualification Inspection - The qualification inspection shall consist of all the tests of 4.8 and the Life Test of 4.5.

4.3.1 Qualification Inspection Samples - Qualification inspection samples shall consist of 3 samples. The samples submitted shall have been previously subjected only to the individual inspections. The samples shall be forwarded, at the contractor's expense, to the laboratory designated in the letter of authorization (Section 6.6). Three samples shall be retained by the contractor for the Qualification when tests are run by the contractor at his plant.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:28Z
Check the source to verify that this is the current version before use.

4.3.1.1 Qualification Inspection Sample Identification - The qualification inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

> Samples for Qualification Inspection
> (Name of Equipment)
> (Submitted by (Manufacturer's name, date and part number) for Qualification Inspection in accordance with Specification MIL-I- (AS) under authorization (reference letter authorizing tests)).

4.3.2 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved qualification samples. Each equipment shall be capable of successfully passing the same tests as imposed on the qualification samples. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.4 Quality Conformance Test - The contractor shall furnish all samples and shall be responsible for accomplishing all tests except Sampling Plan B tests which will be conducted at a government laboratory designated by the procuring activity. Quality conformance test, except for Sampling Plan B, shall be under the supervision of the government quality control representative. The contractor shall furnish test reports showing detailed quantitative results for all tests required by this specification, signed by an authorized representative of the laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Quality Conformance tests shall consist of the following tests:

(a) Individual Tests
(b) Sampling Plan A Tests
(c) Sampling Plan B Tests
(d) Reliability Assurance Tests
(e) Special Tests

4.4.2 Sampling Plans - The Sampling Plans shall consist of Sampling Plan A and Sampling Plan B tests. The tests samples selected for sampling tests shall first have passed the Individual Test. The test samples which have been subjected to Sampling test shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual tests.

Quantity Offered for Acceptance	Quantity to be Selected for Sampling Plan A Tests
First 15	1 (Zero when Sampling
Next 50	1 Plan B is invoked)
Each 100	
of remainder	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:28Z
Check the source to verify that this is the current version before use.

*Page 16, Paragraph 4.4.2.1: In title and first sentence change "sampling tests" to read "Sampling Plan A tests".

* Page 16: Add the following Paragraphs

"4.4.2.2 Sampling Plan B Instructions - One sample shall be selected at random from the first 15 produced on contract and submitted within ten days after manufacture. These samples shall be forwarded at the contractor's expense to a Government laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

Indicator Gyro ID-____/A
Submitted by (Manufacturers name, date) for
Production Acceptance Sampling Plan B Inspection
of MIL-G-81 with Contract No. _____
Manufacturers Part No. _____.

4.4.2.2.1 Sampling Plan B Approval - Approval of Sampling Plan B instruments shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material or performance defect made evident during this test shall be corrected by the Contractor to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of instruments under the contract to cease until proper corrective action is approved and accomplished.

4,4,2,2,2 Sampling Plan B Inspection - Each sample selected for Sampling Plan B testing shall be subjected to the following test

(a) Sampling Plan A Tests
(b) Vibration Failure
(c) Radio Noise Interference
(d) High Temperature Exposure
(e) High Altitude - Low Temperature
(f) Temperature Shock
(g) Humidity
(h) Salt Spray
(i) Fungus Resistance
(j) Impact Shock
(k) Acceleration
(l) Sand and Dust"

*Page 17, Paragraph 4.4.3.1: Delete

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:28Z
Check the source to verify that this is the current version before use.

Page 17, Paragraphs 4.4.3.2, 4.4.3.2.1: Delete and Substitute

"4.4.3.2 Reliability Production Acceptance Phase - Reliability Production Acceptance Testing shall be conducted through a productionbburn-in. Each equipment produced but not submitted to the Reliability Qualification Test shall be operated for 150 operating hours burn-in, and the test data (operating hours and failures) shall be compared to the Reject Criteria of Test Plan V of MIL-STD-781. (Extend the lines as necessary to accommodate the data omitting the truncation lines). Delivery of equipment shall not be held up to await a decision under the test plan. At any time that the current total of test hours and test failures plotted on Test Plan V curves show a Reject Situation, the Procuring Activity shall be notified. The Procuring Activity reserves the right to stop the acceptance of equipment at any time that a Reject Situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a rejection decision. In the event the total of test hours and test failures plotted on Test Plan V lines extended shown an Accept Situation, the test duration on each subsequently produced equipment shall be reduced to 100 operating hours. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. For the Production Acceptance Phase, Test Level F shall be used, except that the temperature extremes shall be +55°C and -44°C, and vibration shall be 0.5G."

* Page 19, Paragraph 4.5, third sentence: Delete "Preproduction" and substitute "Qualification".

* Page 21, Paragraph 4.6, first sentence, delete "Preproduction" and substitute "Qualification".

Page 23, Paragraph 4.8.7: Add

"The test shall be repeated with power removed."

Page 24, Paragraph 4.8.9.2.1, Second Sentence: Delete and Substitute

"The roll and pitch axis shall not change more than 3.5 degrees in a 4 minute period."

Page 24, Paragraph 4.8.9.2.2, Last Sentence: Delete and Substitute

"The roll and pitch indication shall not change more than 3.5 degrees in a 4 minute period."

Page 24: Add

"4.8.9.2.3 Vertical Angle Erection - With the roll axis of the indicator orientated along the vertical and the spin axis orientated towards the geographical north, the gyro shall be aligned to indicate 0 degrees in pitch and roll. The roll and pitch indication shall not change more than 2 degrees in a 32 second period."

Page 25, Paragraph 4.8.12: Delete and Substitute

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:28Z
Check the source to verify that this is the current version before use.

"4.8.12 Rotor Coast Time - The pitch and roll indication shall not change more than 6 degrees in a 9 minute period."

Page 25, Paragraph 4.8.14, First Sentence: Delete and Substitute.

"The indicator shall be properly connected and subjected to the transient variations specified in MIL-STD-704. No damage shall occur to the indicator. The indicator shall then be tested with the input voltage and frequency being varied within the following limits and combinations."

Page 26, Paragraph 4.8.18, After second sentence: Add

"Linear motion may be used instead of circular motion."

* Page 27, Paragraph 4.8.19 Delete and Substitute

"4.8.19 Vibration Failure - The indicator shall be tested in accordance with Part II, Vibration Test of MIL-T-5422 except the following vibration curve shall be substituted in lieu of Curve II of Figure 2: between 5 HZ and 20 Hz a constant double amplitude of 0.10 inch shall be applied, between 20Hz and 350 Hz a constant 2G vibration shall be applied, between 350 Hz and 2,000 Hz a constant 5G vibration shall be applied. Power, including power to the lighting curcuit, shall be applied to the unit during the vibration test. Following exposure to vibration the unit shall meet the individual test requirements."

Page 28: Add

"4.8.28.1 Operating -Rated power shall be applied to the equipment and precessions in pitch and roll resulting from accelerations shall be limited as stated in 3.7.

4.8.28.1.1 Longitudinal Accelerations - The equipment shall meet the requirements of 3.7.1.

4.8.28.1.2 Coordinated Turns -The equipment shall meet the requirements of 3.7.2.

4.8.28.1.3 Uncoordinated Turns -The equipment shall meet the requirements of 3.7.3."

Page 29, Paragraph 4.8.30, Third Sentence: Delete and Substitute

"The error in either pitch or roll shall not exceed three degrees. The error induced by roll maneuvers at a rate of $300^{\circ} \pm 10^{\circ}$ per second shall not exceed one degree. Futhermore, rotating the indicator in either direction at the above rates shall not cause damage to the indicator."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:28Z
Check the source to verify that this is the current version before use.

Page 29, Add

"4.8.31 Sand and Dust - The indicator shall be subjected to Sand and Dust Tests, Procedure I of Specification MIL-E-5272. At the end of sand exposure, the indicator shall not be damaged and shall meet the requirements of the Individual Tests."

Pages 29 and 30, Paragraphs 5.1, 5.2, 5.3, 5.4, and 5.4.1: Delete and Substitute

"5.1 Packaging, Packing, Preservation and Marking - Shall be in accordance with MIL-G-81559."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:28Z
Check the source to verify that this is the current version before use.

*Page 32 Add:

"6.6 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity."

"6.7 Maintainability Demonstration - Maintainability Demonstration shall be in accordance with MIL-STD-471 except as modified herein. If the Demonstration has been performed on a previous contract or order, it need not be performed."

"6.8 Ordering Data - Procurement documents should specify the following:

(a) Title, number, and date of specification.

(b) The quantity, MS part number and type number, if any, of the instrument desired.

(c) Levels of packaging and packing desired.

(d) The laboratory that shall conduct tests.

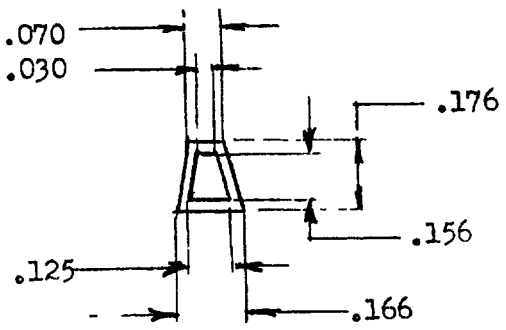
(e) Reliability test phases (See 4.7)."

Page 37, Figure 5: Delete and substitute the new Figure 5 of this Amendment.

Project Number 6610-N051

An asterisk preceding a page number denotes a change to be made to the specification which did not appear in Amendment 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:28Z
Check the source to verify that this is the current version before use.



TOLERANCES FOR DECIMALS ± .005

FIGURE 5 BANK INDEX

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:28Z
Check the source to verify that this is the current version before use.